Exp No:2	# Python Fundamentals
Date:	

AIM

To implement the academic performance analysis of students using Python fundamentals.

ALGORITHM

1. Store student IDs in a tuple.
2. Store student details in a dictionary.
3. Calculate average marks.
4. Determine academic risk level based on marks, attendance, and study hours.
5. Display student performance details.

PROGRAM:

```
# -------------------------------
# Student Performance Analysis
# -------------------------------

# Tuple to store student IDs
student_ids = ('S101', 'S102', 'S103', 'S104')

# Dictionary to store student academic details
students = {
    'S101': {'name': 'Asha', 'assignment': 78, 'test': 80, 'attendance': 92, 'hours': 8},
    'S102': {'name': 'Ravi', 'assignment': 65, 'test': 68, 'attendance': 85, 'hours': 5},
    'S103': {'name': 'Meena', 'assignment': 88, 'test': 90, 'attendance': 96, 'hours': 10},
    'S104': {'name': 'Kiran', 'assignment': 55, 'test': 58, 'attendance': 78, 'hours': 4}
}

# Function to calculate average score
def calculate_average(assignment, test):
    return (assignment + test) / 2

# Function to determine academic risk level
def determine_risk(avg_score, attendance, hours):
    if avg_score < 60 or attendance < 80 or hours < 5:
        return "High Risk"
    elif avg_score < 75:
        return "Moderate Risk"
    else:
        return "Low Risk"
```

OUTPUT:

```
----- Student Performance Report ---

Student ID   : S101
Name         : Asha
Average Score: 79.00
Attendance   : 92%
Study Hours  : 8 hrs/week
Risk Level   : Low Risk
-----------------------------------
Student ID   : S102
Name         : Ravi
Average Score: 66.50
Attendance   : 85%
Study Hours  : 5 hrs/week
Risk Level   : Moderate Risk
-----------------------------------
Student ID   : S103
Name         : Meena
Average Score: 89.00
Attendance   : 96%
Study Hours  : 10 hrs/week
Risk Level   : Low Risk
-----------------------------------
Student ID   : S104
Name         : Kiran
Average Score: 56.50
Attendance   : 78%
Study Hours  : 4 hrs/week
Risk Level   : High Risk
-----------------------------------
PS C:\Users\kavim>
```

```
# Processing student records
print("----- Student Performance Report -----\n")

for sid in student_ids:
    data = students[sid]
    avg = calculate_average(data['assignment'], data['test'])
    risk = determine_risk(avg, data['attendance'], data['hours'])

    print(f"Student ID   : {sid}")
    print(f"Name         : {data['name']}")
    print(f"Average Score: {avg:.2f}")
    print(f"Attendance   : {data['attendance']}%")
    print(f"Study Hours  : {data['hours']} hrs/week")
    print(f"Risk Level   : {risk}")
    print("-" * 35)
```

Python program using tuples, dictionaries, functions, and conditional logic to analyze student academic risk.

Preparation	20	
Implementation	30	
Viva	15	
Record	10	
Total	75	

RESULT

Thus, the academic performance of students was implemented using Python fundamentals and uploaded to GitHub.

GitHub Link : **https://github.com/KavimugilRajasekar/MachineLearningLab.git**